

Econ 131
Fall 2024

Midterm

March 2024

Student Name:

Student ID:

Exam Instructions

- **Explanation should be written using pens (we recommend black or blue ink, as these often scan the best).** No pencils, except for graphs.
- **Do not write your solutions on pages that say “Do not write on this page”.** Answers written on these pages will not be graded.
- **Show your work.** Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below.** For those writing on a non-printed copy, please just write “Academic Honesty Pledge as on exam”, and sign your name.
If you do not affirm this pledge, your exam will be marked invalid.

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: _____

1. True/False/Uncertain Questions (12 points total)

(a) A reduction in the marginal income tax rate should always make people work more.

(b) Once you have established that a treatment group and control group exhibit parallel trends in the period before a tax change, it is valid to estimate the effect of the tax change using the difference-in-differences estimator we studied in class:

$$DD = [Y^{T,After} - Y^{C,After}] - [Y^{T,Before} - Y^{C,Before}]$$

.

(c) The U.S. poverty rate stayed constant over the 1972-2005 period discussed in class.

(d) Suppose that the supply curve for widgets (in a standard graph with price on the y-axis and quantity on the x-axis) has a slope of 1, while the demand curve for widgets has a slope that is -1000. Most of the incidence of a tax on widgets will be on consumers.

- (e) If earnings do not respond to taxes, then a utilitarian government would want to set taxes and transfers in order to equalize everyone's consumption.

- (f) The experience of Denmark indicates that high earners do not decide where to live based on taxes.

2. Income Taxation and Labor Supply 16 points total)

Assume that households have the following utility function over *consumption* and *labor*, given by:

$$U(c, l) = c - \frac{l^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}$$

where c denotes consumption, l denotes hours of labor supply, and $\eta > 0$ is nonzero. Households' only source of earnings is labor income. The hourly wage is w and the price of consumption is 1. The government collects revenue through a "flat tax" at rate t on earnings z : a household with pre-tax earnings z pays tz in taxes. The government also sends households a lump-sum transfer $T > 0$.

- (a) What is the household budget constraint with this tax and transfer system? How does this lump-sum transfer differ from the actual implementation of transfers in the United States? (3 points)

- (b) Solve for the household's optimal consumption c^* and labor supply choice l^* as functions of w, η, t and T . (4 points)

- (c) Compute the taxable income elasticity (i.e. the elasticity of labor supply with respect to post-tax wage). Based on empirical findings discussed in class, is this value usually greater than or less than 1 for top income earners? (3 points)

- (d) What is the elasticity of the labor supply with respect to the lump-sum transfer? Given this elasticity, does the income effect dominate the substitution effect? (*3 point*)
- (e) Let $l^d(w) = w^{-\eta^d}$ be the labor demand of firms. What will be the equilibrium wage w^* as a function of η, η^d , and t ? (*3 points*)

3. Government Budget 16 points total)

Assume the government's only source of revenue each year t is income taxation and its only expenditure is defense spending, denoted by G_t . Let τ_t be the linear tax rate on income in year t , and let $z(\tau) = (1 - \tau)^\phi$ be the income level for a given tax rate. Let the growth rate of the economy be zero $g = 0$.

- (a) For a given tax rate τ_t , what is the tax revenue of the government as a function of τ_t and ϕ ? (2 point)

- (b) What is the primary deficit each period as a function of τ_t, ϕ , and G_t ? (2 point)

- (c) If the government runs a balanced budget (i.e., defense spending equals tax revenue) every year, what is the maximum spending on defense G_t that the government can achieve as a function ϕ ? (4 point)

- (d) If the government DOES NOT run a balanced budget, the government can use debt to finance expenses. Suppose the government begins year t with debt B_t , collects income tax revenue using tax rate τ_t , spends G_t on defense. If the government faces an interest rate r on its debt what will be the debt next period B_{t+1} as a function of B_t, G_t, τ_t, r and ϕ ? (2 point)

- (e) Now suppose that tax instruments are chosen so that the debt is constant across periods (i.e. $B_{t+1} = B_t = B$), taxes are constant across periods (i.e. $\tau_t = \tau$) and defense spending is constant across periods (i.e. $G_t = G$). Using the equation for the evolution of government debt from year to year, what will the sustainable debt level B equal as a function of τ, G, r , and ϕ ? (4 point)
- (f) If the government runs a primary deficit each period and the interest rate is positive (so that $r > g$), can the government sustain a constant positive debt level B ? (2 point)

Reminder: please sign the Academic Honesty Pledge on the first page!